Market Efficiency and the Size Premium

Evidence from US Equity Quintiles, 1926–2025

Fama-French Data Library

Abstract

We examine market efficiency and the size premium in US equities from July 1926 to December 2025 using monthly and daily returns from the Fama-French data library. Autocorrelation tests and Lo-MacKinlay (1988) variance ratios reject the random-walk null across the full sample (Ljung-Box $p < 0.001$). A size gradient is pervasive: small-cap stocks (Lo 20) exhibit $AC(1)$ ⊠ 0.19 monthly and earn 1.26%/month versus 0.07 and 0.94%/month for large-caps (Hi 20). Subperiod analysis reveals that aggregate market predictability collapses post-1976 (Q-p = 0.48) while small-cap predictability persists, consistent with non-synchronous trading and limits to arbitrage.

1. Introduction

The weak-form efficient market hypothesis (EMH) asserts that current prices fully reflect the information in past returns, implying serial independence in price changes (Fama, 1970). One of the most robust findings in empirical finance is the size premium: small-capitalisation stocks have historically earned higher average returns than large-caps (Banz, 1981; Fama and French, 1992). The two sit together naturally: if small-cap returns are more autocorrelated, part of the size premium could be compensation for the frictions that stop arbitrageurs from trading the predictability away.

This paper addresses both questions using a century-long dataset spanning July 1926 to December 2025, comprising 1,194 monthly and 26,151 daily observations. We test for serial dependence via Ljung-Box Q-statistics and Lo-MacKinlay variance ratio tests across multiple horizons, and split the sample at April 1976 to assess whether market efficiency has improved over time.

2. Data

We use monthly and daily total return series from the Fama-French data library. The market return is constructed as Mkt-RF + RF. Size quintile portfolios (Lo 20, Qnt 2-4, Hi 20) are value-weighted, formed annually using NYSE market-capitalisation breakpoints, with Lo 20 containing the smallest 20% of stocks. Returns are in percent; observations coded -99.99 or -999 are treated as missing. The full sample runs from July 1926 to December 2025.

3. Methodology

Autocorrelation and the Ljung-Box test

We estimate the sample autocorrelation function at lags 1 through 5 and test joint significance via the Ljung-Box Q-statistic (Ljung and Box, 1978). Under the null of no serial dependence at any lag up to K, the statistic is asymptotically chi-squared with K degrees of freedom:

$$Q(K) \;=\; n\sum_{k=1}^{K} \hat{\rho}_k^2 \;\sim\; \chi^2(K)$$

where n is the sample size and ρ̂k is the sample autocorrelation at lag k. We set K = 5.

Variance ratio tests

Following Lo and MacKinlay (1988), the variance ratio at horizon q is:

$$\mathrm{VR}(q) \;=\; \frac{\mathrm{Var}\left(r_t^{(q)}\right)}{q \cdot \mathrm{Var}\left(r_t^{(1)}\right)}$$

Under the random-walk null, VR(q) = 1 for all q. VR(q) > 1 implies positive serial correlation (momentum); VR(q) < 1 implies mean reversion. Monthly horizons: q ⊠ {2,4,8,12}; daily: q ⊠ {2,5,10,20}.

4. Empirical Results

The size premium and cumulative wealth

Figure 1 plots cumulative wealth from $1 invested in each size quintile and the aggregate market over 1926–2025 (log scale). The size gradient is immediately striking: Lo 20 compounds to dramatically higher terminal wealth than Hi 20. Table 1 confirms Lo 20 earns a mean monthly return of 1.26% (std 8.79%) versus 0.94% (std 5.10%) for Hi 20, a premium of around 32 bps/month that is monotone across all five quintiles.

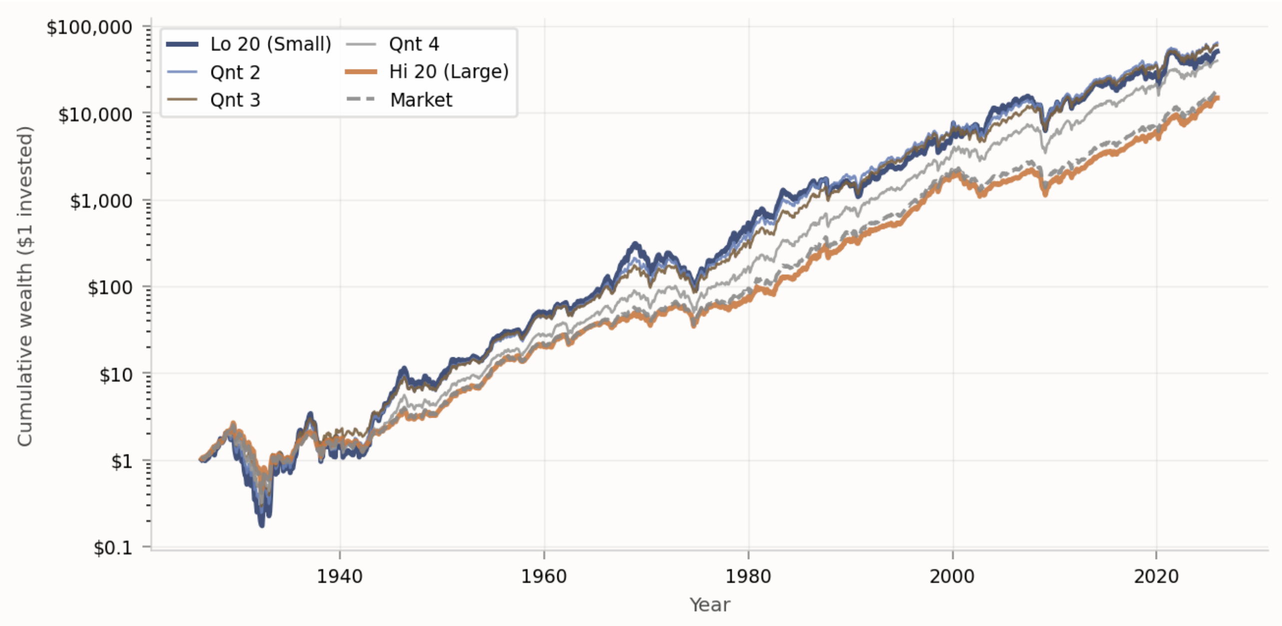

Figure 1. Cumulative wealth from $1 invested in each size quintile and the aggregate market, logarithmic scale. Monthly total returns, Fama-French library, July 1926–December 2025.

Serial correlation and the random-walk null

Tables 1 and 2 report full-sample summary statistics. Every series decisively rejects the random-walk null at the 1% level. Both AC(1) and VR(q) decrease monotonically from Lo 20 to Hi 20 at both frequencies. Monthly AC(1) ranges from 0.186 (Lo 20) down to 0.069 (Hi 20); daily AC(1) spans 0.149 to 0.022. Figure 4 visualises the gradient. The outsized daily autocorrelation in small-caps is consistent with non-synchronous trading (Lo and MacKinlay, 1990): thinly-traded stocks incorporate market-wide information with a lag, mechanically inflating own-autocorrelations.

Table 1. Full-sample summary statistics: Monthly returns

	Mean (%/mo)	Std (%)	AC(1)	AC(2)	AC(3)	Q(5)	p-val	VR(2)	VR(4)	VR(8)	VR(12)
Market	0.9615	5.2973	0.0899	-0.0263	-0.0841	23.14	0.0003***	1.0877	1.0764	1.1065	1.1913
Lo 20	1.2631	8.7859	0.1862	0.0180	-0.0701	56.19	0.0000***	1.1674	1.2512	1.1927	1.2441
Qnt 2	1.2040	7.5724	0.1728	-0.0081	-0.1024	56.04	0.0000***	1.1453	1.1740	1.0626	1.0962
Qnt 3	1.1518	6.8109	0.1417	-0.0152	-0.0962	37.38	0.0000***	1.1200	1.1285	1.0639	1.1063
Qnt 4	1.0774	6.1582	0.1144	-0.0115	-0.0880	26.48	0.0001***	1.1005	1.1077	1.0869	1.1371
Hi 20	0.9375	5.0970	0.0690	-0.0277	-0.0804	21.54	0.0006***	1.0695	1.0498	1.1015	1.2023

*VR(q) = Lo-MacKinlay variance ratio at q-month horizon. Q(5) = Ljung-Box statistic with 5 lags. */**/*** = rejection at 10%/5%/1%.*

Table 2. Full-sample summary statistics: Daily returns

	Mean (%/day)	Std (%)	AC(1)	AC(2)	AC(3)	Q(5)	p-val	VR(2)	VR(5)	VR(10)	VR(20)
Market	0.04287	1.0782	0.0461	-0.0248	-0.0010	77.7	0.0000***	1.0475	1.0556	1.0638	1.1182
Lo 20	0.04980	1.2824	0.1493	0.0350	0.0512	803.0	0.0000***	1.1496	1.3445	1.5549	1.8442
Qnt 2	0.05039	1.2472	0.1287	0.0065	0.0353	496.8	0.0000***	1.1295	1.2590	1.3742	1.5500
Qnt 3	0.04934	1.1910	0.1218	-0.0078	0.0227	426.9	0.0000***	1.1231	1.2193	1.3021	1.4325
Qnt 4	0.04711	1.1437	0.1025	-0.0198	0.0162	307.2	0.0000***	1.1042	1.1704	1.2156	1.3054
Hi 20	0.04271	1.0846	0.0223	-0.0297	-0.0070	39.6	0.0000***	1.0236	1.0041	0.9907	1.0226

VR(q) at q-day horizons {2,5,10,20}.

Variance ratios across horizons

Figure 2 plots VR(q) against horizon for the market, Lo 20, and Hi 20 at both frequencies. All three series display VR $>$ 1 increasing with horizon, confirming positive medium-run momentum. Lo 20 reaches VR(12) $=$ 1.34 monthly and VR(20) $=$ 1.84 daily. Hi 20 remains close to 1.0 throughout.

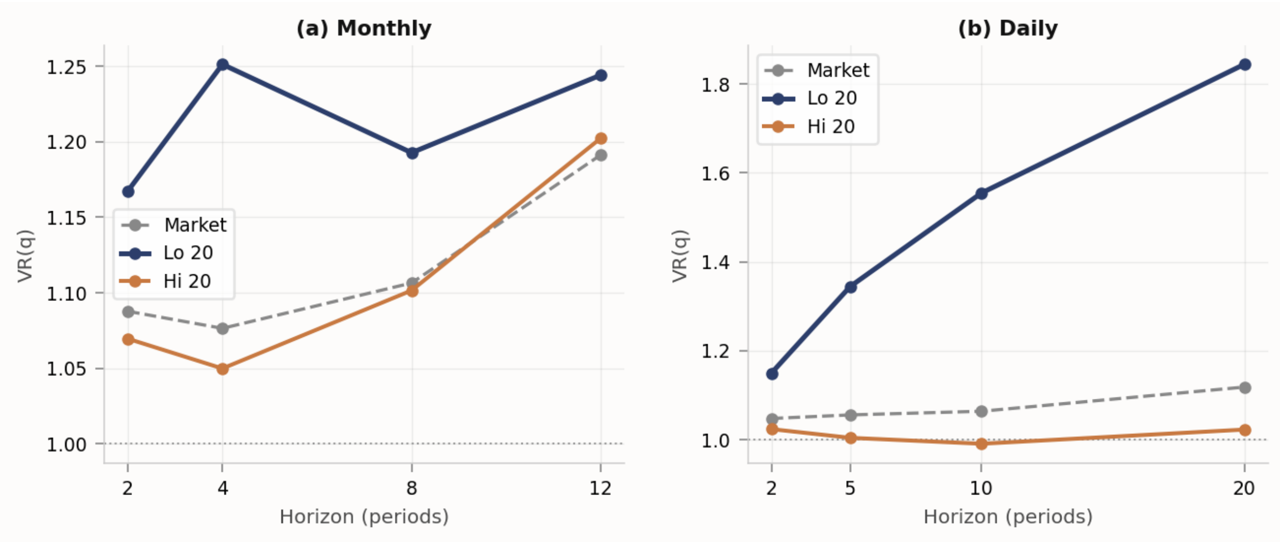

Figure 2. Lo-MacKinlay variance ratios at multiple horizons. Left: monthly (q = 2,4,8,12). Right: daily (q = 2,5,10,20). Dotted line at VR = 1 is the random-walk null.

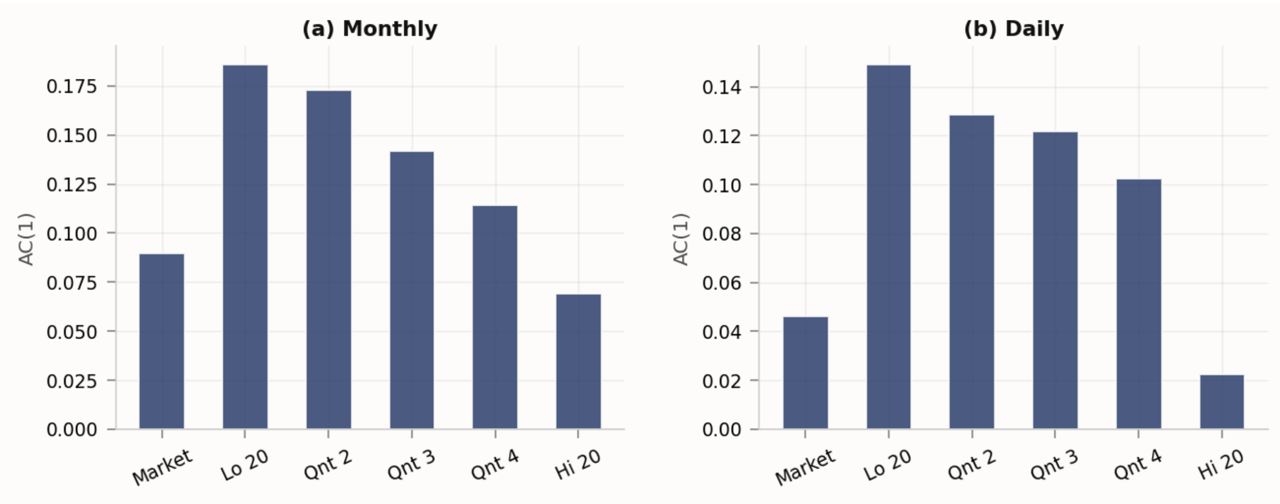

Figure 4. Lag-1 autocorrelation AC(1) across size quintiles and the aggregate market at monthly (left) and daily (right) frequency.

5. Subperiod Analysis

We split the sample at April 1976, broadly coinciding with the growth of index funds, discount brokers, and computerised trading. Figure 3 marks the break with a vertical line and Table 3 reports subperiod statistics.

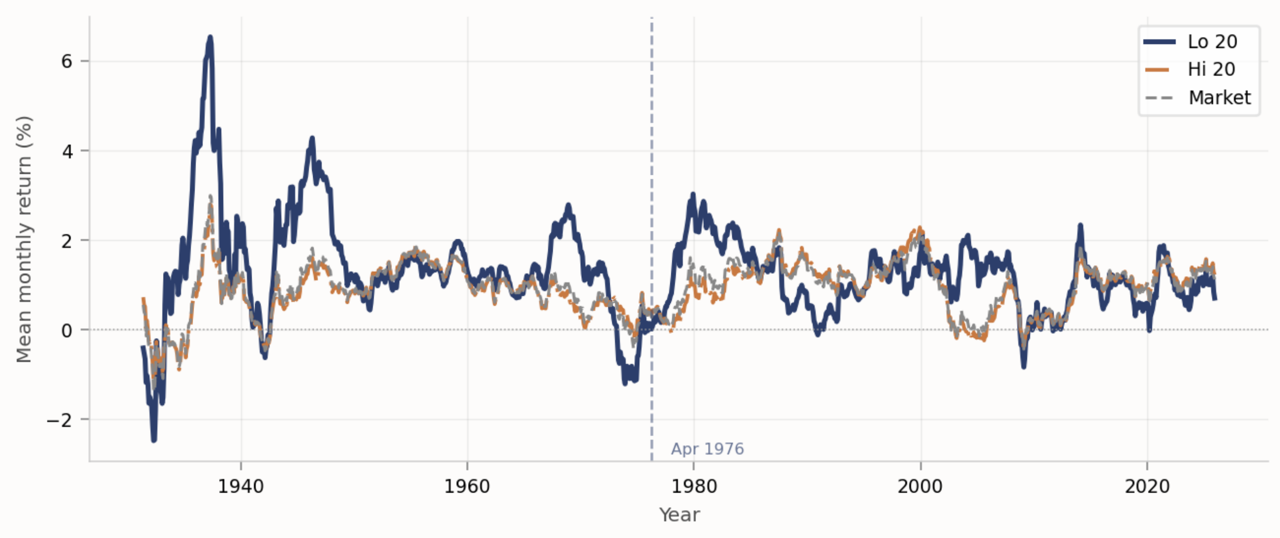

Figure 3. Rolling 60-month mean monthly return (%) for Lo 20, Hi 20, and the market. Vertical line marks the April 1976 subperiod break.

Table 3. Subperiod statistics: Monthly returns

	Mean (%)	Std (%)	AC(1)	Q(5)	p-val	VR(2)	VR(4)	VR(8)	VR(12)
Market	0.877	6.035	0.124	22.28	0.000***	1.117	1.123	1.166	1.299
Lo 20	1.406	10.838	0.200	32.67	0.000***	1.177	1.282	1.235	1.337
Qnt 2	1.243	8.988	0.220	42.10	0.000***	1.184	1.257	1.157	1.257
Qnt 3	1.156	7.996	0.181	29.12	0.000***	1.150	1.191	1.142	1.253
Qnt 4	1.014	7.121	0.152	22.66	0.000***	1.129	1.167	1.166	1.277
Hi 20	0.851	5.790	0.107	22.27	0.000***	1.103	1.095	1.146	1.284
Market	1.046	4.442	0.026	4.51	0.479	1.039	1.001	1.017	1.034
Lo 20	1.120	6.083	0.143	16.87	0.005***	1.150	1.187	1.122	1.076
Qnt 2	1.165	5.831	0.061	8.96	0.111	1.071	1.015	0.895	0.817
Qnt 3	1.148	5.377	0.054	6.37	0.272	1.064	1.013	0.930	0.855
Qnt 4	1.141	5.019	0.038	4.76	0.446	1.051	1.005	0.956	0.911
Hi 20	1.024	4.297	-0.001	6.30	0.278	1.013	0.977	1.039	1.094

*Sub1 = July 1926 – March 1976; Sub2 = April 1976 – December 2025. Thin rule separates subperiods. */**/*** as above.*

The contrast is stark. In Sub1 the market Q-statistic is 22.28 ($p < 0.001$) and $VR(12) = 1.30$. In Sub2 it falls to 4.51 with $p = 0.48$; the aggregate market is statistically indistinguishable from a random walk. Lo 20 retains significant predictability in both subperiods (Sub2 $Q\text{-}p = 0.005$). Hi 20 in Sub2 is a near-perfect random walk ($Q\text{-}p = 0.28$).

Figure 3 shows the time-variation in the size premium. Rolling 5-year mean returns for Lo 20 and Hi 20 alternate in dominance: the premium largely disappeared in the 1990s technology bull market but re-emerged in the early 2000s and 2020s. This fits Lo's (2004) adaptive markets framework: the premium reflects varying risk appetites and competitive conditions rather than a fixed arbitrage opportunity.

6. Robustness Checks and Hypothesis Tests

This section stress-tests the main results five ways: unit root and normality checks, heteroskedasticity-robust variance ratio tests, a non-parametric runs test, a direct test of the size premium, and structural break tests around the 1976 boundary.

6.1 Stationarity and distributional diagnostics

Before running autocorrelation tests it is worth confirming the series are stationary. ADF decisively rejects the unit-root null for all six series (test statistics between -8.5 and -8.9, $p < 0.001$), while KPSS fails to reject the stationarity null at any conventional level. The Jarque-Bera statistics reveal strong departures from normality. Lo 20 exhibits extreme positive skewness (2.54) and excess kurtosis of 25.7. This motivates the use of heteroskedasticity-robust test statistics throughout.

Table 4. Unit root tests (ADF, KPSS) and Jarque-Bera normality — Monthly

	ADF stat	ADF p	KPSS stat	KPSS p	Skewness	Ex. Kurt.	JB stat
Lo 20	-8.481	0.0000***	0.1070	>0.10	2.537	25.666	34,054***
Qnt 2	-8.725	0.0000***	0.0540	>0.10	1.559	16.914	14,717***
Qnt 3	-8.842	0.0000***	0.0420	>0.10	0.943	12.194	7,575***
Qnt 4	-8.906	0.0000***	0.0190	>0.10	0.656	10.592	5,667***
Hi 20	-8.490	0.0000***	0.0470	>0.10	0.094	7.089	2,502***
Market	-8.712	0.0000***	0.0350	>0.10	0.124	7.365	2,701***

ADF null: unit root. KPSS null: stationarity. Ex. Kurt. = excess kurtosis (Fisher). JB significant at 1% throughout.

6.2 Heteroskedasticity-robust variance ratio tests (M2)

The VR statistics in Section 4 assume homoskedastic innovations. Given the heavy tails above, we use the M2 statistic of Lo and MacKinlay (1988), which is robust to conditional heteroskedasticity. Lo 20 stays significant at $q = 2$ ($z = 2.83$, $p = 0.005$) and $q = 4$ ($z = 2.49$, $p = 0.013$). The market is borderline at $q = 2$ ($p = 0.078$). Hi 20 is insignificant throughout. The size gradient holds after the correction.

Table 5. Heteroskedasticity-robust VR tests (M2) — Full sample, monthly

	q=2 VR / z	q=2 p	q=4 VR / z	q=4 p	q=8 VR / z	q=8 p	q=12 VR / z	q=12 p
Lo 20	1.1674 (2.827)	0.0047***	1.2512 (2.489)	0.0128**	1.1927 (1.322)	0.1861	1.2440 (1.338)	0.1810
Qnt 2	1.1452 (2.414)	0.0158**	1.1739 (1.671)	0.0947*	1.0625 (0.412)	0.6804	1.0961 (0.510)	0.6103
Qnt 3	1.1199 (2.104)	0.0354**	1.1285 (1.255)	0.2095	1.0637 (0.416)	0.6773	1.1061 (0.557)	0.5775
Qnt 4	1.1005 (1.860)	0.0629*	1.1076 (1.123)	0.2615	1.0868 (0.606)	0.5444	1.1370 (0.768)	0.4425
Hi 20	1.0694 (1.403)	0.1606	1.0501 (0.546)	0.5847	1.1015 (0.712)	0.4764	1.2023 (1.133)	0.2573
Market	1.0876 (1.764)	0.0778*	1.0766 (0.840)	0.4011	1.1064 (0.754)	0.4506	1.1912 (1.082)	0.2791

*z = Lo-MacKinlay M2 heteroskedasticity-robust statistic, asymptotically N(0,1). */**/*** as above.*

Table 6 breaks the robust VR tests down by subperiod. In Sub2, Lo 20 remains significant at $q = 2$ ($z = 2.98$, $p = 0.003$) and $q = 4$ ($z = 2.01$, $p = 0.044$). The market and Hi 20 are both flat. Small-cap predictability survives the correction.

Table 6. Heteroskedasticity-robust VR by subperiod — Market, Lo 20, Hi 20

	q=2 VR / z	q=2 p	q=4 VR / z	q=4 p	q=8 VR / z	q=8 p	q=12 VR / z	q=12 p
Market Sub1	1.1164 (1.625)	0.1042	1.1242 (0.941)	0.3469	1.1662 (0.813)	0.4159	1.2993 (1.168)	0.2429
Market Sub2	1.0391 (0.732)	0.4640	1.0015 (0.016)	0.9872	1.0172 (0.119)	0.9049	1.0332 (0.187)	0.8519
Lo 20 Sub1	1.1769 (2.218)	0.0265**	1.2844 (2.105)	0.0353**	1.2352 (1.211)	0.2261	1.3370 (1.384)	0.1664
Lo 20 Sub2	1.1496 (2.977)	0.0029***	1.1870 (2.012)	0.0442**	1.1226 (0.867)	0.3862	1.0762 (0.436)	0.6626
Hi 20 Sub1	1.1027 (1.439)	0.1501	1.0958 (0.722)	0.4705	1.1467 (0.709)	0.4781	1.2837 (1.093)	0.2744
Hi 20 Sub2	1.0127 (0.239)	0.8110	0.9776 (-0.235)	0.8139	1.0396 (0.273)	0.7848	1.0923 (0.516)	0.6056

Sub1 = July 1926–March 1976; Sub2 = April 1976–December 2025.

6.3 Non-parametric runs test

As a fully non-parametric cross-check we use the Wald-Wolfowitz runs test, which checks whether the sequence of positive and negative returns is consistent with randomness, with no assumptions about the return distribution. Lo 20 has dramatically fewer runs than expected (529 actual vs 597 expected, $z = -3.94$, $p < 0.001$), as do Qnt 2 and Qnt 3 at the 5% level. Hi 20 and the market are both consistent with independence. The non-parametric evidence lines up cleanly with everything above.

Table 7. Wald-Wolfowitz runs test — Monthly, full sample

	Actual runs	Expected runs	z-stat	p-val
Lo 20	529	597.0	-3.941	0.0001***
Qnt 2	559	596.5	-2.174	0.0297**
Qnt 3	562	598.0	-2.085	0.0371**
Qnt 4	573	598.0	-1.448	0.1477
Hi 20	576	598.0	-1.274	0.2027
Market	594	598.0	-0.232	0.8168

*H_0: return signs are independently drawn. */**/*** as above.*

6.4 Formal test of the size premium

We test H_0: E[Lo20 - Hi20] = 0 using an OLS t-test and a Newey-West (1987) HAC t-test with 12 lags. The mean spread is 0.33%/month (3.91%/year). Both tests reject the null at 10% (OLS: $t = 1.94$, $p = 0.053$; NW: $t = 1.73$, $p = 0.083$). The borderline significance reflects the high variance of the spread series (std around 8.8%/month, inflated by Depression-era observations).

Table 8. t-tests for the size premium (Lo 20 - Hi 20 monthly spread)

Test	Mean spread (%/mo)	Annualised (%/yr)	t-stat	p-val
OLS t-test	0.3256	3.91	1.937	0.053*
Newey-West (12 lags)	0.3256	3.91	1.733	0.083*

*NW = Newey-West HAC standard errors. */**/*** as above.*

6.5 Structural break: mean and variance

We test whether mean returns or variance changed at the 1976 break. Mean returns did not change significantly for either the market ($t = -0.58$, $p = 0.57$) or Lo 20 ($t = 0.55$, $p = 0.59$). Variance did change sharply: market variance fell from 36.4 to 19.8 ($F = 1.84$, $p < 0.001$), and Lo 20 variance fell from 117.3 to 37.1 ($F = 3.17$, $p < 0.001$). This explains the drop in post-1976 Q-statistics. Small-cap autocorrelation did not disappear; the series got less volatile, which mechanically shrinks

$n \times \rho^2$.

Table 9. Structural break tests at April 1976 — Monthly

Series / Test	Sub1	Sub2	Statistic	p-val
Market — Mean (%/mo)	0.874	1.050	t = -0.576	0.565
Market — Variance	36.37	19.76	F = 1.840	0.000***
Lo 20 — Mean (%/mo)	1.402	1.124	t = 0.546	0.586
Lo 20 — Variance	117.28	37.06	F = 3.165	0.000***

*Welch t-test allows unequal variances. F-test: H_0 = equal variances. */**/*** as above.*

7. Discussion

Market efficiency. The full-sample rejection of the random walk is well documented (Lo and MacKinlay, 1988; Poterba and Summers, 1988). The subperiod evidence shows the aggregate market has become genuinely more efficient: post-1976, it passes the weak-form EMH test. Index funds and algorithmic trading have speeded up price adjustment considerably.

Size premium. Small-caps earn meaningfully higher returns (Lo 20 at 1.26%/month vs 0.94% for Hi 20 over the full sample). Part is compensation for illiquidity (Amihud and Mendelson, 1986) and higher systematic risk. Part likely reflects the thin-trading and underreaction effects shown above.

Predictability gradient. The monotone size gradient in AC(1) and VR(q) is the most consistent finding in this paper, across every test and both frequencies. The natural explanation is non-synchronous pricing: small-cap stocks lag behind market-wide information, producing a lead-lag structure that persists even as the aggregate market becomes efficient (Lo and MacKinlay, 1990; Hou, 2007).

8. Conclusion

Using nearly 100 years of Fama-French data, this paper finds two things that work together: aggregate market predictability that has largely faded post-1976, and a size premium with a persistent autocorrelation gradient that has not. The post-1976 market passes the weak-form EMH test; small-cap portfolios do not, in either subperiod. The frictions sustaining small-cap predictability are real and costly to arbitrage away. The size premium is time-varying, which fits an adaptive-markets view better than a fixed risk story.

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